

Exact and Structural Methods for the Job Sequencing and Tool Switching Problem

Mid-term results

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Outline

The problem and the two questions

A worst-case theory of the heuristic

Which structure decides the gap?

The computational study

Remaining work

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The problem

Data. Jobs J , tools T , magazine capacity b ; job j requires $T_j \subseteq T$, $|T_j| \leq b$.

Decision. An order of the jobs and a loading policy; each tool insertion costs 1.

- ▶ Fixed order $\Rightarrow$ loading solved exactly by KTNS (keep tool needed soonest) in $O(n|T|)$ [Tang–Denardo 1988]
- ▶ Joint problem NP-hard [Crama et al. 1994]

Practice: group, then sequence. Partition jobs into fewest magazine-fitting groups (JGP), order the groups (small TSP), run KTNS. Applied verbatim in industry; assessed only *empirically* until now [Burger et al. 2015].

Two questions, one quantity

Q1 (heuristic)

How far can group-then-sequence be from optimal? When is it exact?

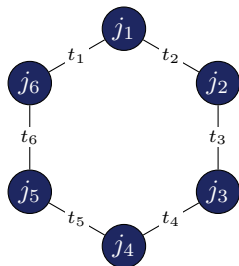
Q2 (exact methods)

Can the SSP be solved through tool *configurations* (column generation, Benders), as the JGP so successfully is?

The organising quantity: the coverage bound $|U| - b$

Every used tool beyond the initial load must be inserted once, so $Z^* \geq |U| - b$. It is the LP value of every natural configuration relaxation (proved) — and on part of the instance space it *is* the optimum.

The smallest counterexample: the 6-ring



$$T_j = \{j, j+1 \bmod 6\}, b = 3$$

- ▶ optimum: $Z^* = 3$ — a *four*-group walk, one tool per transition
- ▶ heuristic ($K^* = 3$ groups, best order): $H = 4$

$$\text{gap} = 1, \quad \text{ratio} = 4/3$$

The grouping phase is to blame: minimal cardinality hides the cheaper grouping. g disjoint copies $\Rightarrow$ gap = g : **unbounded**.

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The gap identified exactly

Theorem (Grouping exactness)

$Z^* = \min_{\mathcal{P}} \gamma(\mathcal{P})$ over all feasible groupings. Hence $\text{gap} = H - Z^* =$ the price of restricting to minimum cardinality.

Proved zero-gap classes (every b): $K^* \leq 2, \quad |U| \leq b + 1, \quad Z^* \leq 2.$

The setup-cost spectrum: SSP $\longleftrightarrow$ JGP

In practice a changeover costs a fixed setup ρ beyond the per-tool swaps.

Theorem (Both ends explicit)

$\rho < \frac{1}{n-1} \Rightarrow$ *the augmented problem is the SSP.*

$\rho > H - Z^* \Rightarrow$ *every optimum uses minimum-cardinality grouping: the heuristic is exactly optimal.*

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- ▶ the transition lives on $[1/(n-1), H - Z^*]$: it is governed by the gap itself,
- ▶ real tooling reports $\rho \approx 3\text{--}60 \Rightarrow$ production sits far on the collapsed (JGP) side — a rigorous licence for standard practice.

The extremal law, quantitatively

For $K^* = 3$ (configurations $\subseteq U$, $q = |U| - b$, $R = Z^* - q$):

$$H = q + \min \min(x_{ab}, x_{bc}, x_{ca}), \quad x_{uv} = |(C_u \cap C_v) \setminus C_w|$$

$$\text{gap} \leq \max\left(0, \min\left(q, \left\lfloor \frac{2b-q}{3} \right\rfloor\right) - R\right) \quad (\text{tight at every known extremum})$$

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Zero-gap classes, every b : $K^* \leq 2, \quad |U| \leq b + 1, \quad Z^* \leq 2.$

Determining the law: a certified elimination

Natural candidate (all data $b \leq 4$): $\text{gap} \leq K^* - 2$.

- ▶ **proved** for every $b \leq 4$ at $K^* = 3$, and outside the corner $2b \geq q + 3R + 6$ in general;
- ▶ the corner is nonempty — and the law **fails** there:

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Certificate ($b = 5$, $K^* = 3$, $|U| = 9$)

$T = (\{0, 1, 2, 4, 7\}, \{0, 4, 5\}, \{1, 5\}, \{2, 6, 8\}, \{3, 4, 5, 8\})$:

$Z^* = 4$, $H = 6$

$\text{gap} = 2 > K^* - 2$, ratio $3/2$ — and it attains our quantitative bound *with equality*.

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Open (posed exactly): is $\max(0, \min(q, \lfloor (2b - q)/3 \rfloor) - R)$ the true worst case at $K^* = 3$?

Status: attained at five (b, q, R) points (two new, at $R = 1$); the $(5, 3)$ and $(6, 3)$ corners resist directed search — the question lives exactly there.

The $4/3$ ratio, and what remains

Theorem ($b = 3$)

Every instance with $K^ \leq 3$ satisfies $H/Z^* \leq 4/3$.*

- ▶ ingredients: gap ≤ 1 at $K^* = 3, b = 3$ (counting law) and positive gap $\Rightarrow Z^* \geq 3$ (small-optimum lemma);
- ▶ the case $K^* \geq 4$ is the *one* surviving conjecture — populated (witnesses with ratio $5/4$ exist), no violation known;
- ▶ the edge-family pattern “positive gap $\Rightarrow K^* = 3$ ” does *not* extend — found and corrected this month.

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Which structure decides these questions?

Negative (gap side). Two instances with the *same* circuit clutter, same K^* , same Z^* have different gaps (0 vs 1) $\Rightarrow$ no job-compatibility invariant can decide exactness; tool-level overlaps are essential.

Theorem (Covering side: odd rings, via blocking duality)

Clusters over maximal groups; blocker = minimal covers (K^ = its min size). On the k -ring, $b = 3$: fractional cover = $k/2$, $K^* = \lceil k/2 \rceil$ — **tight iff k even.***

Odd circular structure: first provable obstruction for the set-covering relaxation behind JGP branch-and-price — the parallel of odd holes.

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The computational instrument

Why run it. The theorems single out $|U| - b$; the benchmark measures it across the standard suites. And a prior question: the SSP's polynomial loading subproblem is the textbook precondition for Benders — untested until now.

- ▶ methods: 8 Benders configs (ours), LSS '04, Catanzaro-F4 '15, SSPMF '24, our branch-and-price (PCF', PTF)
- ▶ 1,421 standard instances ($n \leq 40$, $|T| \leq 60$); 3600 s / 600 s; pinned threads; early stop after 8 straight failures
- ▶ one canonical metric: every returned sequence re-scored by KTNS
- ▶ integrity: 806 instances solved by someone — **zero** cross-method disagreements

Results

	primary (411)	secondary (1,010)
SSPMF (2024)	138 /165	(not run, by design)
LSS (2004, modern CPLEX)	91/98	668 /678
Catanzaro F4 (2015)	91/99	455/490
best Benders (ours)	1/9	26/48
PCF' B&P (ours, $n \leq 25$)		94/379 pooled
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- ▶ SSPMF strongest on primary — the published SOTA claim *replicates*
- ▶ 2004 formulation + modern solver $\approx$ 2015 formulation, and dominates secondary: solver generation outweighs formulation vintage
- ▶ everyone's wall: $n = 15$ with $|T| \geq 30$; nobody solves $n \geq 30$

The diagnosis: bound-limited, mechanically

Our Benders on capacity-loose instances: optimum found in seconds; the *proof* bound frozen at its trivial root value for the full hour ($\sim 250k$ nodes, $\sim 75k$ cuts). Fractional cuts: **structurally inactive** — no violated cut exists at fractional orderings (verified in controlled runs).

The measured split

Among the 806 solved instances: 47.9% have $Z^* = |U| - b$ exactly (upper-biased estimate). The suites split in half between the regime where heuristic and exact methods win at the root, and the regime where new mathematics is required.

PCF'/PTF confirm the same law from the branch-and-price side (15% of PCF' optima at the root; PTF's stronger bound is the one lever that exceeds $|U| - b$).

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Programme

1. **Implemented, cluster re-run queued:** the Benders master with the coverage row $\theta \geq |U|$ — verified; on bound-tight instances the proof now closes *at the root, with zero cuts* (the 6-ring: 0 cuts vs. 443 before).
2. **Theory (open):** the exact extremal law at $K^* = 3$ (is the counting bound the truth?); the $4/3$ case $K^* \geq 4$.
3. **With Prof. Wagler:** PTF polytope facets on the known witnesses (PORTA); the odd-ring idealness question for JGP.

Questions for today: which side of the blocking pair (clusters/covers vs. circuits/groups) to pursue first; priority between the Benders re-run and the polytope study; and whether the remaining theory items match your expectations for the closing period.